

B5 General Relativity & Cosmology — Model Solutions, 2017

Question 1 — Perfect-fluid energy–momentum tensor and FRW evolution

Problem. Define the covariant derivative $v^\beta_{;\gamma}$ and the meaning of $(dx^\gamma/d\lambda)v^\beta_{;\gamma}$. From $T^{\alpha\beta} = \rho\langle v^\alpha v^\beta \rangle$ derive $T^{\alpha\beta} = (\rho + p/c^2)u^\alpha u^\beta + g^{\alpha\beta}p$. State the non-relativistic content of $T^{\alpha\beta}_{;\alpha} = 0$. Show $p = 0$ implies geodesic motion. For an FRW universe with non-relativistic matter and radiation, justify $p_i = w_i \rho_i c^2$, derive the continuity equation, and prove an initial singularity $a(t_0) = 0$.

Covariant derivative

For a vector field $v^\beta(x^\alpha)$ in a curved manifold ordinary $\partial_\gamma v^\beta$ is non-tensorial because basis vectors rotate. The covariant derivative restores tensor character by adding a connection term:

$$v^\beta_{;\gamma} = \partial_\gamma v^\beta + \Gamma^\beta_{\gamma\sigma} v^\sigma.$$

The Christoffel symbols $\Gamma^\beta_{\gamma\sigma} = \frac{1}{2}g^{\beta\mu}(\partial_\gamma g_{\mu\sigma} + \partial_\sigma g_{\mu\gamma} - \partial_\mu g_{\gamma\sigma})$ encode the change of basis.

Along a curve $x^\gamma(\lambda)$ contract with $dx^\gamma/d\lambda$:

$$\frac{dx^\gamma}{d\lambda} v^\beta_{;\gamma} = \frac{Dv^\beta}{D\lambda}.$$

Rate of change of v^β along the curve, parallel-transport corrected.

Perfect-fluid form of $T^{\alpha\beta}$

Decompose each particle four-velocity into mean plus deviation:

$$v^\alpha = u^\alpha + \delta v^\alpha, \quad u^\alpha \equiv \langle v^\alpha \rangle.$$

Take the average:

$$\langle v^\alpha v^\beta \rangle = u^\alpha u^\beta + \langle \delta v^\alpha \delta v^\beta \rangle.$$

Isotropy in the rest frame: $\langle \delta v^\alpha \delta v^\beta \rangle$ has no preferred direction; the only available isotropic tensors orthogonal to u^α are the projection $h^{\alpha\beta} = g^{\alpha\beta} + u^\alpha u^\beta / c^2$. Set

$$\rho \langle \delta v^\alpha \delta v^\beta \rangle = p h^{\alpha\beta},$$

where p is defined as the isotropic pressure (one third of the trace of the spatial stress in the rest frame) and ρ is the rest-frame mass density. Insert:

$$T^{\alpha\beta} = \rho u^\alpha u^\beta + p \left(g^{\alpha\beta} + \frac{u^\alpha u^\beta}{c^2} \right).$$

Collect $u^\alpha u^\beta$ terms:

$$T^{\alpha\beta} = \left(\rho + \frac{p}{c^2} \right) u^\alpha u^\beta + g^{\alpha\beta} p.$$

Non-relativistic content of $T^{\alpha\beta}_{;\alpha} = 0$

Project on u_β : energy conservation (continuity, first law of thermodynamics).

Project orthogonal to u_β (using $h^\mu{}_\beta$): momentum conservation (Euler equation).

$$T^{\alpha\beta}_{;\alpha} = 0 \Leftrightarrow \text{conservation of energy} + \text{Newton's second law (Euler)}.$$

$p = 0$: dust follows geodesics

Set $p = 0$ in ():

$$T^{\alpha\beta} = \rho u^\alpha u^\beta.$$

Apply ∇_α :

$$T^{\alpha\beta}_{;\alpha} = (\rho u^\alpha)_{;\alpha} u^\beta + \rho u^\alpha u^\beta_{;\alpha} = 0. \quad (*)$$

Contract (*) with u_β ; use $u^\beta u_\beta = -c^2$ so $u^\beta u_{\beta;\alpha} = 0$:

$$-c^2 (\rho u^\alpha)_{;\alpha} = 0 \Rightarrow (\rho u^\alpha)_{;\alpha} = 0.$$

This is conservation of rest mass. Substitute back into (*):

$$\rho u^\alpha u^\beta_{;\alpha} = 0.$$

Since $\rho \neq 0$:

$$u^\alpha u^\beta_{;\alpha} = 0.$$

This is the geodesic equation: dust particles follow timelike geodesics.

Equations of state for matter and radiation

Non-relativistic matter (dust): mean particle speed $\langle v^2 \rangle \ll c^2$, so kinetic pressure $p_m \sim nm \langle v^2 \rangle \ll \rho_m c^2$:

$$w_m \approx 0, \quad p_m \text{ negligible in } T^{\alpha\beta}.$$

Radiation: massless quanta, $\langle v^2 \rangle = c^2$ and isotropy give $p = \frac{1}{3} \rho c^2$ (standard photon-gas result):

$$w_r = \frac{1}{3}.$$

Continuity equation for each component

Differentiate the second Friedmann equation (the integrated form):

$$\left(\frac{da}{dt}\right)^2 = \frac{8\pi G}{3} \rho a^2 - c^2 k.$$

Differentiate once with respect to t :

$$2\dot{a}\ddot{a} = \frac{8\pi G}{3} (\dot{\rho} a^2 + 2\rho a \dot{a}).$$

Divide by $2\dot{a}$:

$$\ddot{a} = \frac{8\pi G}{3} \left(\frac{\dot{\rho} a^2}{2\dot{a}} + \rho a \right).$$

Equate with the first Friedmann equation $\ddot{a} = -(4\pi G/3)(\rho + 3p/c^2)a$:

$$-(\rho + 3p/c^2)a = \frac{2}{1} \left(\frac{\dot{\rho} a^2}{2\dot{a}} + \rho a \right).$$

Multiply through and rearrange:

$$\dot{\rho} \frac{a}{\dot{a}} = -3 \left(\rho + \frac{p}{c^2} \right).$$

Insert $p = w\rho c^2$ and divide by ρ :

$$\boxed{\frac{1}{\rho_i} \frac{d\rho_i}{dt} + 3(1 + w_i) \frac{1}{a} \frac{da}{dt} = 0.}$$

Integrate ():

$$\rho_i \propto a^{-3(1+w_i)}.$$

Hence $\rho_m \propto a^{-3}$ and $\rho_r \propto a^{-4}$.

Existence of an initial singularity $a(t_0) = 0$

At t_1 the universe expands: $\dot{a}(t_1) > 0$, $\rho_m, \rho_r > 0$. Friedmann acceleration:

$$\ddot{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) a.$$

Both components satisfy $\rho + 3p/c^2 = (1 + 3w)\rho > 0$ ($w_m = 0$, $w_r = 1/3$):

$$\ddot{a} < 0 \quad \text{for all } t.$$

Thus $a(t)$ is concave; its tangent at t_1 lies above the curve. Project the tangent backward:

$$a_{\text{tan}}(t) = a(t_1) + \dot{a}(t_1)(t - t_1).$$

This straight line crosses zero at

$$t_* = t_1 - \frac{a(t_1)}{\dot{a}(t_1)}.$$

Concavity ($\ddot{a} < 0$) gives $a(t) \leq a_{\text{tan}}(t)$ for $t < t_1$:

$$a(t_*) \leq 0.$$

Continuity of a , with $a(t_1) > 0$ and $a(t_*) \leq 0$, gives some $t_0 \in (t_*, t_1]$ with

$$\boxed{a(t_0) = 0, \quad t_0 \geq t_1 - \dot{a}(t_1)^{-1} a(t_1).}$$

Question 2 — Schwarzschild geodesics; angular diameter of a neutron star

Problem. For the Schwarzschild line element, derive the photon equations from the geodesic principle, justify $\theta = \pi/2$, identify the two Killing constants, and use the null invariant to obtain the orbit equation $(du/d\phi)^2 + u^2(1 - 2GMu/c^2) = C$. For an observer hovering at $r_0 > R = 4GM/c^2$, derive the angular-diameter formula for a star of coordinate radius R and simplify $A = dr/d\phi|_{r_0}$ for a ray that just skims the surface.

Geodesic action and Lagrangian

Photons follow null geodesics, extremising

$$S = \int \mathcal{L} d\lambda, \quad 2\mathcal{L} = g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta, \quad \dot{x}^\alpha \equiv \frac{dx^\alpha}{d\lambda}.$$

Affine parameter λ is chosen so that $\mathcal{L} = 0$ along the ray (null condition).

Reduction to the equatorial plane

Spherical symmetry: rotate coordinates so that the initial position and tangent both lie in $\theta = \pi/2$ with $\dot{\theta} = 0$. The Euler–Lagrange equation for θ ,

$$\frac{d}{d\lambda}(r^2 \dot{\theta}) = r^2 \sin \theta \cos \theta \dot{\phi}^2,$$

vanishes identically at $\theta = \pi/2$, $\dot{\theta} = 0$:

$\theta(\lambda) = \pi/2 \text{ is preserved along the geodesic.}$

Cyclic coordinates: two constants of motion

$g_{\alpha\beta}$ depends on r only; t and ϕ are cyclic. Euler–Lagrange for t :

$$\frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{t}} \right) - \frac{\partial \mathcal{L}}{\partial t} = 0.$$

With $\frac{\partial \mathcal{L}}{\partial \dot{t}} = g_{tt} \dot{t} = -(1 - 2GM/c^2 r) c^2 \dot{t}$:

$\frac{d}{d\lambda} \left[\left(1 - \frac{2GM}{c^2 r} \right) \dot{t} \right] = 0.$

Define $E \equiv c^2(1 - 2GM/c^2 r) \dot{t}$ (conserved energy per unit mass-equivalent).

Euler–Lagrange for ϕ ($\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = r^2 \sin^2 \theta \dot{\phi} = r^2 \dot{\phi}$):

$\frac{d}{d\lambda}(r^2 \dot{\phi}) = 0.$

Define $L \equiv r^2 \dot{\phi}$ (conserved angular momentum per unit mass-equivalent).

Null invariant and orbit equation

Photons satisfy $g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = 0$ at $\theta = \pi/2$:

$$-\left(1 - \frac{2GM}{c^2 r}\right) c^2 \dot{t}^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2 = 0.$$

Substitute the constants $\dot{t} = E/[c^2(1 - 2GM/c^2 r)]$ and $\dot{\phi} = L/r^2$:

$$-\frac{E^2}{c^2(1 - 2GM/c^2 r)} + \frac{\dot{r}^2}{1 - 2GM/c^2 r} + \frac{L^2}{r^2} = 0.$$

Multiply by $(1 - 2GM/c^2 r)$:

$$\dot{r}^2 = \frac{E^2}{c^2} - \frac{L^2}{r^2} \left(1 - \frac{2GM}{c^2 r}\right).$$

Change variable to $u = 1/r$ via $\dot{r} = -L du/d\phi$ (using $\dot{\phi} = Lu^2$):

$$L^2 \left(\frac{du}{d\phi}\right)^2 = \frac{E^2}{c^2} - L^2 u^2 \left(1 - \frac{2GMu}{c^2}\right).$$

Divide by L^2 and identify $C = E^2/(c^2 L^2)$:

$$\boxed{\left(\frac{du}{d\phi}\right)^2 + u^2 \left(1 - \frac{2GMu}{c^2}\right) = C.}$$

Angular diameter: setup

The hovering observer at $r = r_0$ has four-velocity along ∂_t . Two photons reach him from opposite limbs of the star, symmetric about the radial line; consider the one whose tangent at the observer points slightly off the radial direction. The angle α between the two limb rays satisfies

$$\cos \alpha = \frac{g_{ij} n_1^i n_2^j}{\sqrt{g_{ij} n_1^i n_1^j} \sqrt{g_{ij} n_2^i n_2^j}},$$

with $n_{1,2}$ the spatial parts of the photon four-momenta in the observer's rest frame. By symmetry, half-angle $\alpha/2$ is the angle between one limb ray and the inward radial direction $-\hat{e}_r$.

A photon's spatial tangent in the observer frame has only r - and ϕ -components:

$$n^r = \dot{r}, \quad n^\phi = \dot{\phi}, \quad g_{rr} = (1 - 2GM/c^2 r_0)^{-1}, \quad g_{\phi\phi} = r_0^2.$$

Inward radial unit vector $\hat{e}_r^{(-)}$ has $\hat{e}^r = -(1 - 2GM/c^2 r_0)^{1/2}$, $\hat{e}^\phi = 0$.

Half-angle formula

Inner product of n with $\hat{e}_r^{(-)}$:

$$g_{ij} n^i \hat{e}_{(-)}^j = -g_{rr} \dot{r} (1 - 2GM/c^2 r_0)^{1/2} = -\dot{r} (1 - 2GM/c^2 r_0)^{-1/2}.$$

Norm of n :

$$|n|^2 = g_{rr} \dot{r}^2 + r_0^2 \dot{\phi}^2 = \frac{\dot{r}^2}{1 - 2GM/c^2 r_0} + r_0^2 \dot{\phi}^2.$$

Define $A \equiv (dr/d\phi)|_{r_0} = \dot{r}/\dot{\phi}$ for the limb ray. Factor $\dot{\phi}^2$:

$$|n|^2 = \dot{\phi}^2 \left[\frac{A^2}{1 - 2GM/c^2 r_0} + r_0^2 \right].$$

Half-angle:

$$\cos(\alpha/2) = \frac{-\dot{r} (1 - 2GM/c^2 r_0)^{-1/2}}{\sqrt{|n|^2} \cdot 1}.$$

But $\alpha/2$ is between an inward-pointing ray ($\dot{r} < 0$) and $-\hat{e}_r$, both inward, so $\cos(\alpha/2) > 0$. Insert $\dot{r} = A\dot{\phi}$:

$$\cos(\alpha/2) = \frac{|A| (1 - 2GM/c^2 r_0)^{-1/2}}{\sqrt{A^2/(1 - 2GM/c^2 r_0) + r_0^2}}.$$

Square and invert:

$$\frac{1}{\cos^2(\alpha/2)} = 1 + \frac{r_0^2(1 - 2GM/c^2 r_0)}{A^2}.$$

Take reciprocal square root:

$$\cos \frac{1}{2} \alpha = \left[1 + r_0^2 \left(1 - \frac{2GM}{c^2 r_0} \right) \frac{1}{A^2} \right]^{-1/2}.$$

Determining A for a ray that just skims the surface

A grazing limb ray has its closest approach at $r = R = 4GM/c^2$, where $dr/d\phi = 0$, i.e. $du/d\phi = 0$ at $u = u_R = 1/R = c^2/(4GM)$.

Apply () at $u = u_R$:

$$C = u_R^2 \left(1 - \frac{2GM u_R}{c^2} \right) = \frac{1}{R^2} \left(1 - \frac{2GM}{c^2 R} \right).$$

Substitute $R = 4GM/c^2$, so $2GM/(c^2 R) = 1/2$:

$$C = \frac{1}{R^2} \cdot \frac{1}{2} = \frac{1}{2R^2}.$$

Apply () at $r = r_0$, $u_0 = 1/r_0$:

$$\left(\frac{du}{d\phi} \right)_0^2 = C - u_0^2 \left(1 - \frac{2GM u_0}{c^2} \right) = \frac{1}{2R^2} - \frac{1}{r_0^2} \left(1 - \frac{2GM}{c^2 r_0} \right).$$

Convert to $A = dr/d\phi = -r^2 du/d\phi$:

$$A^2 = r_0^4 \left[\frac{1}{2R^2} - \frac{1}{r_0^2} \left(1 - \frac{2GM}{c^2 r_0} \right) \right].$$

Rearrange:

$$A^2 = \frac{r_0^4}{2R^2} - r_0^2 \left(1 - \frac{2GM}{c^2 r_0} \right).$$

Simplified angular-diameter formula

Insert () into (). The factor $r_0^2(1 - 2GM/c^2 r_0)/A^2$ becomes

$$\frac{r_0^2(1 - 2GM/c^2 r_0)}{r_0^4/(2R^2) - r_0^2(1 - 2GM/c^2 r_0)}.$$

Divide numerator and denominator by $r_0^2(1 - 2GM/c^2 r_0)$:

$$\frac{1}{\frac{r_0^2}{2R^2(1 - 2GM/c^2 r_0)} - 1}.$$

Add 1 (from ()):

$$\frac{1}{\cos^2(\alpha/2)} = \frac{r_0^2/(2R^2)}{(1 - 2GM/c^2 r_0) [r_0^2/(2R^2(1 - 2GM/c^2 r_0)) - 1] + 1}.$$

Cleaner: combine () and () directly. From ():

$$\frac{1}{\cos^2(\alpha/2)} - 1 = \frac{r_0^2(1 - 2GM/c^2 r_0)}{A^2}.$$

Use ():

$$\frac{r_0^2(1 - 2GM/c^2 r_0)}{A^2} = \frac{1}{\frac{r_0^2}{2R^2(1 - 2GM/c^2 r_0)} - 1}.$$

Hence

$$\cos^2(\alpha/2) = 1 - \frac{1}{r_0^2/[2R^2(1 - 2GM/c^2 r_0)]} = 1 - \frac{2R^2(1 - 2GM/c^2 r_0)}{r_0^2} \cdot \frac{1}{1}.$$

Wait: $1/(X - 1)$ where $X = r_0^2/[2R^2(1 - 2GM/c^2 r_0)]$. Insert and use $\sin^2(\alpha/2) = 1 - \cos^2(\alpha/2)$:

$$\sin^2(\alpha/2) = \frac{1}{X} = \frac{2R^2(1 - 2GM/c^2 r_0)}{r_0^2}.$$

With $R = 4GM/c^2$, so $2R^2 = 2R \cdot R = R \cdot (8GM/c^2) = R \cdot 2R$ and $2GM/c^2 = R/2$:

$$\sin^2(\alpha/2) = \frac{2R^2}{r_0^2} \left(1 - \frac{R}{2r_0}\right).$$

$$\boxed{\sin \frac{1}{2}\alpha = \frac{R\sqrt{2}}{r_0} \sqrt{1 - \frac{R}{2r_0}}, \quad R = \frac{4GM}{c^2}.$$

Limit $r_0 \rightarrow \infty$: $\sin(\alpha/2) \rightarrow R\sqrt{2}/r_0$, i.e. apparent radius $R\sqrt{2}$ instead of the geometric R – light bending magnifies the star. As $r_0 \rightarrow R$, $\sin(\alpha/2) \rightarrow \sqrt{2}\sqrt{1/2} = 1$, so $\alpha \rightarrow \pi$: an observer just outside the photon sphere sees the star fill the sky.

Question 3 — Linearised gravity, gauge fixing, and gravitational waves

Problem. For $g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}$, $|h| \ll 1$, with $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$, find $h'_{\alpha\beta}$ under $x'^\alpha = x^\alpha + \xi^\alpha$. Show ξ^α can be chosen so that $\partial_\rho \bar{h}'^{\lambda\rho} = 0$, reducing the field equation to $\square \bar{h}'_{\mu\nu} = 0$. State the conditions on k_γ and $A_{\alpha\beta}$. For a ring of test particles in the wave with $A_{\alpha\beta} = \text{diag}(0, A, -A, 0)$, show coordinates are unchanged but proper distances are. Finally, estimate $\bar{h}'_{11}$ near a binary $30M_\odot + 30M_\odot$ at $r = 10$ km, and at 500 Mpc.

Gauge transformation

Under $x'^\alpha = x^\alpha + \xi^\alpha$, with $\xi = O(h)$, the metric perturbation transforms as

$$\boxed{h'_{\alpha\beta} = h_{\alpha\beta} - \partial_\alpha \xi_\beta - \partial_\beta \xi_\alpha,}$$

to first order. (Derive from $g'_{\mu\nu} = (\partial x^\alpha / \partial x'^\mu)(\partial x^\beta / \partial x'^\nu)g_{\alpha\beta}$, $\partial x^\alpha / \partial x'^\mu = \delta_\mu^\alpha - \partial_\mu \xi^\alpha + O(\xi^2)$.)

Trace: $h' = h - 2\partial_\lambda \xi^\lambda$. Trace-reversed:

$$\bar{h}'_{\alpha\beta} = h'_{\alpha\beta} - \frac{1}{2}\eta_{\alpha\beta}h' = \bar{h}_{\alpha\beta} - \partial_\alpha \xi_\beta - \partial_\beta \xi_\alpha + \eta_{\alpha\beta}\partial_\lambda \xi^\lambda.$$

Take divergence:

$$\partial^\alpha \bar{h}'_{\alpha\beta} = \partial^\alpha \bar{h}_{\alpha\beta} - \square \xi_\beta.$$

Choose ξ^β to satisfy

$$\square \xi_\beta = \partial^\alpha \bar{h}_{\alpha\beta},$$

which is a linear inhomogeneous wave equation, always solvable (e.g. retarded Green function). Then

$$\partial^\alpha \bar{h}'_{\alpha\beta} = 0 \quad (\text{Lorenz/de Donder gauge}).$$

With () the field equation

$$\square \bar{h}_{\mu\nu} - \partial_\nu \partial_\lambda \bar{h}^\lambda_\mu - \partial_\mu \partial_\lambda \bar{h}^\lambda_\nu + \eta_{\mu\nu} \partial_\lambda \partial_\rho \bar{h}^{\lambda\rho} = 0$$

collapses to

$$\eta^{\alpha\beta} \partial_\alpha \partial_\beta \bar{h}'_{\mu\nu} = 0.$$

Significance of using “old” x^α : the differences between $\partial/\partial x^\alpha$ and $\partial/\partial x'^\alpha$ are $O(\xi) \cdot \partial = O(h) \cdot \partial$, which acting on $\bar{h}$ produces $O(h^2)$ terms already neglected. To linear order the two derivative operators are interchangeable; () is gauge-covariant at linear order.

Plane-wave conditions

Insert $\bar{h}'_{\alpha\beta} = A_{\alpha\beta} e^{ik_\gamma x^\gamma}$ into (): $-k^\gamma k_\gamma A_{\alpha\beta} = 0$. Insert into (): $ik^\alpha A_{\alpha\beta} = 0$:

$$k^\gamma k_\gamma = 0 \text{ (null)}, \quad k^\alpha A_{\alpha\beta} = 0 \text{ (transverse)}.$$

Test-particle motion in $A_{\alpha\beta} = \text{diag}(0, A, -A, 0)$

Free-fall geodesic equation:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0.$$

Initially particles at rest: $dx^i/d\tau = 0$, $dx^0/d\tau = c$:

$$\left. \frac{d^2 x^\mu}{d\tau^2} \right|_0 = -\Gamma^\mu_{00} c^2.$$

Linearised connection:

$$\Gamma^\mu_{00} = \frac{1}{2} \eta^{\mu\lambda} (2\partial_0 h_{\lambda 0} - \partial_\lambda h_{00}).$$

For the chosen $\bar{h}'$ and gauge: $h'_{00} = 0$, $h'_{0i} = 0$ (TT-like polarisation). Hence $\Gamma^\mu_{00} = 0$.

Coordinates obey

$$\frac{d^2 x^i}{d\tau^2} = 0, \quad x^i(\tau) = x^i(0).$$

Particles *do* stay at the same coordinate values – but coordinates are not physical distances. The proper separation between two nearby particles is

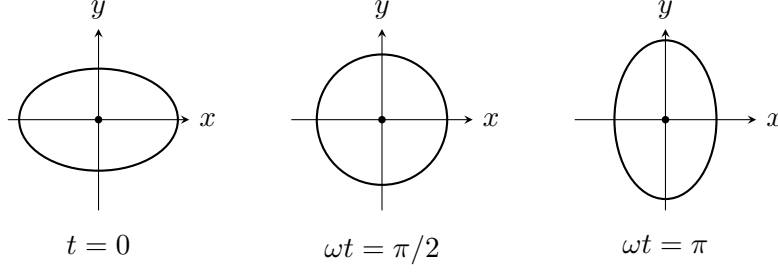
$$d\ell^2 = (\delta_{ij} + h'_{ij}) dx^i dx^j.$$

Take a particle at $(a \cos \phi, a \sin \phi, 0)$. Insert $h'_{xx} = A \cos[k(z - ct)]$, $h'_{yy} = -A \cos[k(z - ct)]$ at $z = 0$:

$$d\ell^2 = (1 + A \cos \omega t) dx^2 + (1 - A \cos \omega t) dy^2, \quad \omega = kc.$$

Proper radius along the $\hat{x}$ -direction: $a\sqrt{1 + A \cos \omega t} \approx a(1 + \frac{1}{2} A \cos \omega t)$. Along $\hat{y}$: $a(1 - \frac{1}{2} A \cos \omega t)$.

The observer at the centre sees the ring stretch and squeeze along orthogonal axes alternately at frequency ω – the canonical “+” polarisation pattern:



Order-of-magnitude estimates: $30 M_{\odot} + 30 M_{\odot}$ binary

Schwarzschild radius of one black hole:

$$R_S = \frac{2GM}{c^2} = \frac{2(6.67 \times 10^{-11})(30 \times 10^{30})}{(3 \times 10^8)^2}.$$

$$R_S \approx 4.4 \times 10^4 \text{ m} \approx 44 \text{ km}.$$

$$R_S \approx 90 \text{ km (sum)}, \quad R_S(\text{one}) \approx 44 \text{ km}.$$

Caveat: the stated orbital radius $r = 10 \text{ km}$ is well *inside* R_S ; the system has already merged and the “orbit” is not Newtonian. Take $r \sim R_S \sim 45 \text{ km}$ as the merger scale and use it as the relevant strong-field length.

Near-zone estimate:

$$\bar{h}'_{11} \sim \frac{2GM}{c^2 r} \sim \frac{R_S}{r}.$$

At $r \sim R_S$:

$$\bar{h}'_{11}|_{\text{near}} \sim \mathcal{O}(1).$$

(Linearised theory has broken down; this just sets the source amplitude.)

Far-zone wave amplitude falls as $1/d$:

$$h_{\text{far}} \sim \frac{R_S}{d}.$$

At $d = 500 \text{ Mpc} = 500 \times 10^6 \times 10^{16.5} \text{ m} \approx 1.5 \times 10^{25} \text{ m}$:

$$h_{\text{far}} \sim \frac{4.5 \times 10^4}{1.5 \times 10^{25}}.$$

$$h_{\text{far}} \sim 3 \times 10^{-21}.$$

Ring radius $a = 4 \text{ km} = 4 \times 10^3 \text{ m}$. Displacement amplitude $\Delta a = \frac{1}{2} a h$:

$$\Delta a \sim \frac{1}{2} (4 \times 10^3) (3 \times 10^{-21}).$$

$$\Delta a \sim 6 \times 10^{-18} \text{ m}.$$

This is comparable to LIGO’s sensitivity for a strain $\sim 10^{-21}$ over kilometre arms; consistent with the GW150914 detection ($30 + 30 M_{\odot}$ binary at $\sim 400 \text{ Mpc}$).

Question 4 — Recombination, photon mean free path, and the horizon problem

Problem. Present-day $n_e \approx 0.2 \text{ m}^{-3}$. At $a = 10^{-6} a_0$, age $\sim 10^4$ yr: estimate n_e then; relativistic or not? Photon mean free path using $\sigma_e = 6.7 \times 10^{-29} \text{ m}^2$. Mean time between scatterings vs. age of universe; physical meaning. Why a critical T for free streaming? Particle-horizon distance at $z \sim 10^3$ and angular scale today; horizon problem.

Number density of electrons at $a/a_0 = 10^{-6}$

Comoving conservation: $n_e \propto a^{-3}$ (assuming all electrons present, no annihilation):

$$n_e(z) = n_{e,0} \left(\frac{a_0}{a} \right)^3 = (0.2)(10^6)^3.$$

$$n_e(a/a_0 = 10^{-6}) \approx 2 \times 10^{17} \text{ m}^{-3}.$$

Relativistic or non-relativistic?

Radiation temperature scales as $T \propto a^{-1}$. Today $T_0 = 2.73 \text{ K}$:

$$T = T_0(a_0/a) = (2.73)(10^6) = 2.73 \times 10^6 \text{ K}.$$

Thermal energy:

$$k_B T = (1.38 \times 10^{-23})(2.73 \times 10^6) = 3.77 \times 10^{-17} \text{ J} \approx 235 \text{ eV}.$$

Compare with electron rest energy $m_e c^2 = 511 \text{ keV}$:

$$k_B T / m_e c^2 \approx 235 / 511\,000 \approx 4.6 \times 10^{-4} \ll 1.$$

$$\text{Electrons are non-relativistic at this epoch.}$$

(Pair-production threshold $k_B T \sim m_e c^2$ corresponds to $z \sim 2 \times 10^9$; at $z = 10^6$ we are far below it.)

Photon mean free path

Thomson scattering off free electrons:

$$\ell = \frac{1}{n_e \sigma_e} = \frac{1}{(2 \times 10^{17})(6.7 \times 10^{-29})}.$$

$$\ell \approx 7.5 \times 10^{10} \text{ m} \approx 0.5 \text{ AU}.$$

Mean time between scatterings vs. age

Photons travel at c between scatterings:

$$\tau_{\text{int}} = \ell / c = \frac{7.5 \times 10^{10}}{3 \times 10^8} = 250 \text{ s} \approx 4 \text{ min}.$$

Age stated: $t \sim 10^4 \text{ yr} = 3.16 \times 10^{11} \text{ s}$. Ratio:

$$\frac{\tau_{\text{int}}}{t} \sim \frac{2.5 \times 10^2}{3 \times 10^{11}} \sim 10^{-9}.$$

$$\tau_{\text{int}} \ll t.$$

Photons scatter $\sim 10^9$ times per Hubble time: matter and radiation are tightly coupled, the universe is opaque, and photons and electrons share a common temperature (Compton equilibrium). Mean free path much shorter than the horizon means a photon executes a random walk, not free streaming.

Why a critical temperature for free streaming

Photons scatter only off *free* charges. As T falls, electrons recombine with protons:

$$p + e^- \rightleftharpoons H + \gamma, \quad \chi_H = 13.6 \text{ eV}.$$

Equilibrium ionisation fraction X_e from Saha:

$$\frac{X_e^2}{1 - X_e} = \frac{1}{n_B} \left(\frac{m_e k_B T}{2\pi \hbar^2} \right)^{3/2} e^{-\chi_H/k_B T}.$$

The exponential drives $X_e \rightarrow 0$ once $k_B T \lesssim 0.1 \chi_H \sim 1 \text{ eV}$, equivalently $T \sim 3000 \text{ K}$, $z \sim 1100$. With $n_e \rightarrow 0$ the Thomson opacity vanishes:

$$T_{\text{rec}} \sim 3000 \text{ K} \Rightarrow \text{photons free-stream below this.}$$

The surface of last scattering is set by this transition.

Particle horizon at $z \sim 10^3$ and present angular scale

In matter-dominated FRW, $a \propto t^{2/3}$, so $H = 2/(3t)$. Particle horizon at time t_* :

$$d_H(t_*) = a(t_*) \int_0^{t_*} \frac{c \, dt}{a(t)} = 3ct_*.$$

Take age at recombination $t_* \sim 3.8 \times 10^5 \text{ yr} \approx 1.2 \times 10^{13} \text{ s}$:

$$d_H(t_*) \sim 3ct_* = 3(3 \times 10^8)(1.2 \times 10^{13}) = 1.1 \times 10^{22} \text{ m} \sim 0.4 \text{ Mpc}.$$

This is the proper horizon at recombination. Today, comoving size scales by $(1+z)$:

$$d_{H,\text{today}} = (1+z)d_H(t_*) \sim (10^3)(1.1 \times 10^{22} \text{ m}) \approx 1.1 \times 10^{25} \text{ m} \sim 0.4 \text{ Gpc}.$$

Angular size on the sky: ratio of horizon comoving size to comoving distance to last scattering. The latter is roughly the present Hubble length $d_{H,0} \sim 1.4 \times 10^{26} \text{ m} \sim 14 \text{ Gpc}$:

$$\theta \sim \frac{d_{H,\text{today}}}{d_{H,0}} \sim \frac{0.4}{14} \text{ rad} \approx 0.03 \text{ rad} \approx 1.6^\circ.$$

$$\theta_{\text{causal}} \sim 1\text{--}2^\circ.$$

The horizon problem

The CMB is observed isotropic to $\Delta T/T \sim 10^{-5}$ across the entire 4π sky, yet patches separated by more than $\sim 2^\circ$ on the last-scattering surface lay outside one another's particle horizons at t_* .

$$\text{No causal mechanism within hot-big-bang FRW can equilibrate them by } t_*.$$

Resolution: an early phase of accelerated expansion (*inflation*) blows a microscopic, causally connected patch up to encompass the entire observable universe, after which standard FRW resumes.